

Split-zero inversion and exact arithmetic capacity control

A source-level reassessment and a complete residual-27 certificate

The Clankers

15 September 2026 — research continuation

Abstract

We return to the distinction between absent zero and supported zero rather than substituting a new meaning of mixed support. A sparse split group-algebra inverse has a different identity from its ordinary amplitude inverse; the precise completion and its lost mask are computed. Actual positive divisor factors instead admit finite, weight-preserving formulas for all exponent multiplicities. At residual 27 a complete finite classification gives 174 minimal capacity profiles and an optimal three-residue-class witness reduction. A separate hand proof classifies all multiplicities in a two-residue family. The same original coefficients have an absolutely convergent Dirichlet carrier and explicit L -function component formulas. No global ES, RH, historical priority or formal-proof certification is claimed.

1 Research assessment and the scalar distinction

The immediate goal is positive integral occupancy of the original Erdős–Straus divisor atlas, not the construction of another object that encodes occupancy after its answer is known. The preceding complexes built from $D_\alpha = R/\gcd(R, g_\alpha)$ do retain failed states, their incoming relations, and their supported zeros. Their formal contraction estimates, however, also hold for arbitrary prescribed arrays of failed defects. Without an additional arithmetic statement, those estimates do not force a successful state. The observation-Gram and prime-indexed Koszul constructions also used “support” differently from the owner’s original scalar distinction.

Here A is a coefficient ring and

$$G(A) = \{\tau\} \sqcup A^\bullet, \quad e = 0_A^\bullet \neq \tau. \quad (\text{S1})$$

The original source distinguishes an empty face, a supported zero amplitude, and an arbitrary supported amplitude. Its mixed-support monad permits a previously absent coordinate to be present with residue zero; the source’s ordinary residue map still has $\{\tau\}$, not a supported infinitesimal fibre, over τ [2, §6.1–6.2]. Its actual internal quotient retains the receiving support and the original boundary primitive [1, §1–4]. We use these distinctions, not a new three-element scalar system.

2 Which inverses preserve absent and supported zero

Let Γ be a finite group and A a nonzero commutative unital ring. A coefficient vector in $G(A)[\Gamma]$ is exactly a pair

$$(S, f), \quad S \subseteq \Gamma, \quad f \in A[\Gamma], \quad \text{nz}(f) \subseteq S. \quad (\text{S2})$$

A coefficient outside S is τ , and a coefficient in S whose amplitude is zero is e . The complete inverse to (S2) reads those two cases separately. Addition and convolution are

$$(S, f) + (T, g) = (S \cup T, f + g), \quad (S, f)(T, g) = (ST, fg). \quad (\text{S3})$$

Indeed a product coefficient is present precisely when some present pair contributes to it. Such contributions can cancel in amplitude but cannot become absent. The global multiplicative identity is $(\{1_\Gamma\}, [1_\Gamma])$; the global additive identity is $(\emptyset, 0)$. Forgetting S is an amplitude homomorphism with complete fibre $\{S : S \supseteq \text{nz}(f)\}$. It is not injective.

Theorem 2.1 (Sparse split units and the nonnegative inverse obstruction). *The global units of $G(A)[\Gamma]$ are exactly the monomials $a^\bullet[g]$ with $a \in A^\times$, $g \in \Gamma$. Moreover, if an ordinary real group-algebra element and its inverse both have nonnegative coefficients, both are monomials.*

Proof. If $(S, f)(T, g)$ is the global identity, $ST = \{1_\Gamma\}$ and neither set is empty. Fixing $t \in T$, injectivity of $s \mapsto st$ forces $|S| = 1$; then $|T| = 1$. Their amplitudes multiply to one. This also proves the converse. For ordinary nonnegative coefficients, the nonzero coefficient support of a product is exactly the product of the two nonzero supports; the same argument applies. $\square$

This theorem concerns the declared global free coefficient semiring. It does *not* prohibit ordinary linear inverses in a fixed full-support fibre of the SplitZero reconstruction. Those inverses have different domains and identities.

An exact six-cycle comparison. The ordinary target in the earlier source is $D_k = [3] + 2[3 - k]$ on C_6 ; its Fourier transform is everywhere nonzero [3, Abstract]. At $k = 2$, with additive indices modulo six, put

$$D = [3] + 2[1], \quad B = \frac{[3] - 2[1] + 4[5]}{9}.$$

Ordinary multiplication gives $DB = [0]$. With its sparse supported lift,

$$\boxed{DB = 1^\bullet[0] + e[2] + e[4] = E_H, \quad H = \{0, 2, 4\}.} \quad (\text{S4})$$

The six individual products give coefficients 1, 0, 0 at 0, 2, 4 respectively; all three sites were used. Thus $E_H^2 = E_H$, but E_H is not the global identity. Also

$$E_H D = [3] + 2[1] + e[5] = D^{\text{sat}}, \quad E_H D^{\text{sat}} = D^{\text{sat}}, \quad D^{\text{sat}} B = E_H. \quad (\text{S5})$$

The target has an inverse relative to the displayed saturated identity after retaining the new present-zero site. Its ordinary Fourier-unit assertion was correct; it did not prove an inverse with the original sparse presence mask. For any subgroup H , $(H, [1])$ gives this same idempotent completion: it preserves amplitudes and replaces S by HS . Keeping the old mask gives the inverse as a decorated correspondence; forgetting it loses absence data. Activation with amplitude zero is not an arithmetic witness.

Fourier observation. In an ordinary finite Fourier matrix and its inverse every entry is nonzero. Under split matrix evaluation, any nonempty input presence mask therefore becomes the full output mask, and inverse Fourier evaluation returns full presence with the original amplitudes. This follows entry by entry from (S3) and the ordinary Fourier inverse. In particular the composite is support completion, not the identity on sparse split vectors. On a predeclared full-support fibre the usual inverse is valid. No new Fourier theorem is claimed; the change of carrier is the point.

3 Original arithmetic before the answer is encoded

Fix $R \equiv 3 \pmod{4}$, $a > R/2$, $\gcd(a, R) = 1$, and put

$$n = 4a - R > R, \quad a = \prod_{i=1}^k q_i^{e_i}, \quad g_i = q_i \pmod{R}. \quad (\text{A1})$$

Primality of n is not needed for the following constructions. The prime applications separately check it. The primes q_i are distinct and ordered, and their multiplicities are never replaced by a generated subgroup. Let $\Gamma_R = (\mathbb{Z}/R\mathbb{Z})^\times$. Define

$$P_{g,e}^E = \sum_{f=0}^{2e} [g]^f, \quad P_{g,e}^M = \sum_{\beta=-e}^e [g]^\beta, \quad t_E = -4^{-1}, \quad t_M = -1 \quad \text{in } \Gamma_R. \quad (\text{A2})$$

Then the coefficient of t_c in $\prod_i P_{g_i, e_i}^c$ is exactly the number of original channel- c states. In E, $u = \prod q_i^{f_i}$; in M, $u = a \prod q_i^{\beta_i}$. The target conditions are respectively $R \mid 4u + 1$ and $R \mid u + a$. This explains why the two targets in (A2) are constant when R is fixed, without deleting the original a .

All coefficients in (A2) are nonnegative integer counts. Sending zero count to τ and positive count m to $m^\bullet$ is a semiring homomorphism from $\mathbb{N}[\Gamma_R]$ to $G(\mathbb{Q})[\Gamma_R]$, since no cancellation or nontrivial zero divisor is possible. This is a declared sparse counting carrier. Declaring every site present instead gives another valid carrier, with the added zero sites retained. Neither is confused with evaluation of a *given* gate: its zero residue is the supported scalar e . Thus an absent target, a signed cancellation, and a successful gate evaluation are different observations with different fibres.

Exact positive denominators. For a target word, let $d = \gcd(a, u)$ and

$$h = d^2/u, \quad r = u/d, \quad s = a/d.$$

At a prime q with valuations $v_q(a) = e$, $v_q(u) = f$, the valuations of r, s, h are $(f - e)^+$, $(e - f)^+$, $e - |f - e|$. They are nonnegative, so $a = hrs$, $u = hr^2$, and $\gcd(r, s) = 1$. The ordered return is

$$E : \kappa = (nr + s)/R, \quad (a, hsk, nhr\kappa), \quad M : \lambda = (r + s)/R, \quad (a, nhs\lambda, nhr\lambda). \quad (\text{A3})$$

Reduction modulo R and $4a \equiv n$ show that the displayed quotients are positive integers. Multiplying the reciprocal sum by $nhrsk$ gives $n\kappa + nr + s = (n + R)\kappa = 4hrs\kappa$ in E; the middle computation gives $n\lambda + r + s = (n + R)\lambda = 4hrs\lambda$. Hence both solve ES. For ordered denominators (a, Y, Z) the inverses are

$$u = \frac{a^2}{RY - na} (E), \quad u = \frac{na^2}{RY - na} (M). \quad (\text{A4})$$

The channel, order, and complete valuation vector remain part of the return. The Type I/II framework is inherited [5, §2, Propositions 2.2 and 2.6].

4 Exact infinite exponent arithmetic with positive coefficients

The next construction starts from the prime factors in (A1), not from the already computed defect or success projector.

Theorem 4.1 (Positive rational carriers and full coefficient fibres). *In $\mathbb{Q}[\Gamma_R][[z]]$,*

$$\boxed{\sum_{e \geq 0} P_{g,e}^E z^e = \frac{1 + z[g]}{(1 - z)(1 - z[g]^2)}, \quad \sum_{e \geq 0} P_{g,e}^M z^e = \frac{1 + z}{(1 - z[g])(1 - z[g]^{-1})}.} \quad (\text{P1})$$

Both right sides have specified positive geometric expansions. Each coefficient has a bijection with the original bounded exponent choices.

Proof. For E expand the two geometric series and the two-term numerator. Their indices $(j, k, \varepsilon) \in \mathbb{N}^2 \times \{0, 1\}$ give $e = j + k + \varepsilon$, $f = 2k + \varepsilon$. Conversely $\varepsilon = f \bmod 2$, $k = (f - \varepsilon)/2$, $j = e - k - \varepsilon$. These are nonnegative exactly on $0 \leq f \leq 2e$. For M the corresponding map is $e = j + k + \varepsilon$, $\beta = j - k$. Its inverse is $\varepsilon = (e - \beta) \bmod 2$, $j = (e - \varepsilon + \beta)/2$, $k = (e - \varepsilon - \beta)/2$; nonnegativity is exactly $|\beta| \leq e$. $\square$

The rational equalities are in the ordinary formal group algebra. The *positive expansions* have canonical sparse split lifts. The denominator expressions are not asserted to have global inverses in $G(\mathbb{Q})[\Gamma_R][[z]]$. This distinction is precisely (S4), now at infinite formal degree. Every coefficient is a finite sum with its original labelled terms, so no divergent summation or choice of analytic branch is involved.

Theorem 4.2 (Periodic count formula and exact finite capacity reduction). *Let $o = \text{ord}_{\Gamma_R}(g)$ and write $e = r + ok$, $0 \leq r < o$. For either channel,*

$$\boxed{P_{g,e}^c = P_{g,r}^c + 2k \sum_{h \in \langle g \rangle} [h].} \quad (\text{P2})$$

In particular its coefficient support is unchanged by replacing e with $\min(e, \lfloor o/2 \rfloor)$. That capacity is the least one at which the whole subgroup is filled. Counts are not unchanged by this support reduction.

Proof. Increasing e by o adds $2o$ consecutive exponents in E. In M it adds o exponents at each end. Each added block of length o traverses the subgroup once, proving (P2) by iteration. A consecutive interval of length $2e + 1$ covers the cyclic subgroup exactly once its length reaches o . Below $\lfloor o/2 \rfloor$ that length is smaller than o . Nonnegative convolution preserves equality of supports, so the reduction holds independently in every factor of a product. $\square$

Thus every fixed finite residue alphabet has an explicit multiaffine quasipolynomial count on each residue class of its exponent parameters modulo their orders: multiply the linear expressions (P2). This formula keeps arbitrarily large multiplicities with finitely many group coordinates.

If several *distinct* original primes have the same residue g , their coefficient supports combine as one interval with capacity $E_g = \sum_{q \equiv g} e_q$. Their counts instead retain

$$C_g(T) = \prod_{q \equiv g} (1 + T + \cdots + T^{2e_q}) \quad (\text{P3})$$

in E, and the analogous centered polynomial in M. The full inverse fibre of aggregate exponent f is all (f_q) with $0 \leq f_q \leq 2e_q$ and $\sum f_q = f$. Equivalently choose the coordinates successively and retain the interval bounds imposed by the remaining capacities. These fibres are nonempty for every $0 \leq f \leq 2E_g$, by induction on the number of intervals. For two factors the fibre length is $\min(2e_1, f) - \max(0, f - 2e_2) + 1$. One prime of multiplicity two has five choices; two distinct primes of multiplicity one have nine. Their residue supports agree but their original amplitudes do not.

5 Complete local classification at residual 27

Here $\Gamma_{27} = \langle 2 \rangle$ has order 18. The two target logarithms are 7 in E and 9 in M. For $1 \leq j \leq 17$ put

$$E_j(a) = \sum_{q \equiv 2^j \pmod{27}} v_q(a), \quad c_j = \left\lfloor \frac{18/\gcd(j, 18)}{2} \right\rfloor. \quad (\text{C1})$$

The identity-residue primes do not alter target existence; their coefficient polynomials and factors in a remain retained.

Theorem 5.1 (Complete finite conductor at $R = 27$). *There are exactly 174 coordinatewise minimal nonnegative capacity profiles for which E or M has a target. They comprise 9 one-class profiles, 99 two-class profiles and 66 three-class profiles. No minimal profile uses more classes. Their maximum total capacity is nine.*

For every $a > 27/2$ coprime to 27, a channel hit exists if and only if its profile $(E_j(a))$ dominates one of those 174 profiles. Consequently any hit has a witness obtained by reserving at most nine actual prime occurrences in a , belonging to at most three nonidentity residue classes modulo 27. All surplus factors remain in the original a and ordered reconstruction.

Proof, including the finite certificate boundary. Equation (P3) identifies coefficient support with the aggregate interval model. Equation (P2) reduces every capacity to $0 \leq E_j \leq c_j$, preserving target existence. Occupancy is increasing in every capacity. These reductions apply to every exponent, not just a computed range.

The following complete finite calculation identifies its minimal occupied profiles. Its full output and target words are in `catalogue27.json`. First enumerate all $2^{17} = 131072$ zero/one supports. A support S has 18-bit masks $V_E(S), V_M(S)$, with the empty mask $\{0\}$. If $j \in S$ is removed and W is the preceding mask, its update is

$$V_E = W \cup (W + j) \cup (W + 2j), \quad V_M = W \cup (W + j) \cup (W - j), \quad (\text{C2})$$

with residues modulo 18. Test membership of 7 and 9. This leaves 360 misses, 256 with only even logarithms and 104 with at least one odd logarithm. A profile supported on even logarithms can never reach either odd target at any multiplicity. For each of the other 104 supports enumerate its positive capacities $1 \leq E_j \leq c_j$, a total of 14040 profiles, using the same interval updates. Keep an occupied profile precisely when all its immediate coordinate predecessors miss. Together with the 85 squarefree minimal profiles this gives 174, with arity counts 9, 99, 66 and maximum capacity sum nine. Both the masks and the exact finite loop bounds are reproduced by the standalone verifier; each retained profile includes a target word and every immediate predecessor is separately checked.

An independent checker uses original unit residues, not logarithmic bit masks. It starts at the zero profile and explores every unoccupied profile in the capped grid. It visits 62812 such profiles and 629841 upward edges. Every occupied boundary edge (567030 in total) dominates a listed profile whose witness and minimality have already been checked in original residues. Thus every monotone path either remains unoccupied or first crosses a checked occupied boundary. This independently verifies the complete antichain, rather than merely corroborating selected entries. It is a finite computer-assisted proof of the displayed small classification; its general reduction was proved before enumeration.

Finally, for a dominated minimal profile reserve its indicated number of actual occurrences in each residue class. Allocate its target exponent to those primes within their reserved capacities, using the interval fibres (P3); this is always possible. E uses the resulting raw exponents. M uses those centered exponents, with every unreserved prime at its original central exponent. Equations (A3)–(A4) give the original marked integer witness. $\square$

A short counterfactual consequence. The 104 squarefree misses containing an odd logarithm have arity distribution 7, 45, 43, 9 at arities 1, 2, 3, 4. Hence, if the prime divisors of a occupy at least five distinct nonidentity residue classes modulo 27 and at least one is $2 \bmod 3$, the residual-27 shell necessarily has a hit. This criterion is pointwise and uses only the given factorization, not an already-found divisor. If such an odd-logarithm shell misses and has exactly four nonidentity classes, its class set must be one of

$$\begin{aligned} &\{5, 10, 19, 23\}, \{5, 8, 11, 13\}, \{8, 11, 13, 25\}, \\ &\{2, 4, 7, 17\}, \{5, 10, 14, 19\}, \{10, 14, 19, 23\}, \\ &\{2, 4, 14, 22\}, \{2, 4, 14, 17\}, \{2, 4, 7, 14\}. \end{aligned}$$

Further multiplicity restrictions are exactly the 174-profile avoidance test; these nine support sets alone are necessary, not sufficient, for failure. The alternative with every prime divisor 1 mod 3 may contain arbitrarily many occurrences and occupies only even logarithms, so it genuinely misses this fixed residual. No global contradiction is claimed from this alternative.

The nine-occurrence bound is sharp for the local residue-profile model: for the sole residue $23 = 2^{11}$, the E word requires raw exponent 17 mod 18, and M requires centered exponent 9 mod 18. Both first become available at capacity nine. No assertion that the corresponding minimal prime instance has been found is needed.

Three residue classes really are necessary at a hard prime. At

$$p = 19489 \equiv 169 \pmod{840}, \quad a = 4879 = 7 \cdot 17 \cdot 41, \quad R = 27,$$

the complete 54-state shell has one hit:

$$\begin{aligned} E : \quad u &= 580601 = 7^2 17^2 41, \quad (h, r, s, \kappa) = (41, 119, 1, 85896), \\ (X, Y, Z) &= (4879, 3521736, 8167578435576). \end{aligned} \tag{C3}$$

Every proper class subbudget misses both targets. The centred vector of this E witness is $(1, 1, 0)$; the third class nevertheless occurs in its *raw* exterior target word. The arity statement refers to the declared E-raw/M-centered budget model, not to the number of nonzero entries in every alternative chart. Trial division through 139 proves primality. The catalogue, residues and ordered inverse are all checked exactly.

6 A hand-verifiable infinite-family subtheorem

Theorem 6.1 (All multiplicities for the 1, 7, 11 residue family). *Suppose every prime factor of a is 1, 7, or 11 modulo 27. Let $A = \sum_{q \equiv 7} v_q(a)$ and $B = \sum_{q \equiv 11} v_q(a)$. Then*

$$E \neq \emptyset \iff B \geq 7 \vee (A \geq 1, B \geq 3) \vee (A \geq 2, B \geq 1), \tag{T1}$$

$$M \neq \emptyset \iff B \geq 9 \vee (A \geq 1, B \geq 5) \vee (A \geq 2, B \geq 1). \tag{T2}$$

Additional residue classes preserve sufficiency of these patterns but not necessity. No assumption of primality for $n = 4a - 27$ is required.

Proof. The logarithms of 7 and 11 to base 2 are $-2, -5$ modulo 18. E is therefore $2i + 5j \equiv 11 \pmod{18}$ with $0 \leq i \leq 2A, 0 \leq j \leq 2B$. If $B = 0$, parity excludes the target. If $A = 0, j \equiv 13 \pmod{18}$, so $B \geq 7$. For $A = 1$, the values $i = 0, 1, 2$ require least nonnegative $j = 13, 9, 5$; hence $B \geq 3$. If $A \geq 2, (i, j) = (3, 1)$ suffices as soon as $B \geq 1$. These exhaust all cases.

M is $2r + 5s \equiv 9 \pmod{18}$ with $|r| \leq A, |s| \leq B$. Again $B = 0$ fails. At $A = 0$ the least possible $|s|$ is 9. At $A = 1$ the three possibilities $r = -1, 0, 1$ have minimum possible $|s|$ respectively 5, 9, 5. At $A \geq 2, (r, s) = (2, 1)$ suffices when $B \geq 1$. The bounded composition fibres (P3) restore actual distinct prime factors; residue 1 factors do not change these congruences. $\square$

The known empty shell $p = 3361, a = 847 = 7 \cdot 11^2$ has $(A, B) = (1, 2)$. Its failure is an inherited regression, not a new ES counterexample [4, §2]. The theorem explains its exact threshold failure. Adding one actual factor 7 gives

$$a' = 5929 = 7^2 11^2, \quad p' = 23689 \equiv 169 \pmod{840}.$$

Its three hits include

$$M : \quad u = 11, \quad (h, r, s, \lambda) = (11, 1, 539, 20), \quad (X, Y, Z) = (5929, 2809041620, 5211580). \tag{T3}$$

Primality is checked by all possible divisors through 153. This is an actual arithmetic extension, not an abstract activation of e . For $a' = ta$, $n' = t(n + R) - R$ and $\gcd(t, R) = 1$, the old E state embeds by $u' = u$ and the old M state by $u' = tu$. Both retain their exact denominator defect, since their gates are respectively unchanged and multiplied by t . The inverse on the marked image is immediate from t . All three new hits lie outside these old images. This does not prove a same-prime descent: n' and n are explicitly different.

As a check of genuinely unbounded multiplicity, for the symbolic $a = 7^{10^{12}} 11^{10^{12}+1}$, (P2) calculates without expanding a or its box the one-channel mass and counts

mass	4000000000008000000000003
E	222222222222555555555555
M	222222222222777777777778.

No primality claim for the corresponding enormous n is made.

7 An exact infinite Dirichlet carrier for the same arithmetic

For fixed R , define $P_c(a) = \prod_{q^e \parallel a} P_{q \bmod R, e}^c$ for $(a, R) = 1$, and

$$\mathcal{F}_c(s) = \sum_{(a, R)=1} P_c(a) a^{-s}.$$

These are locally finite formal Dirichlet series first. For $\sigma = \Re s > 1$ they converge absolutely in the group-algebra coefficient ℓ^1 norm, since

$$\sum_{(a, R)=1} \tau(a^2) a^{-\sigma} = \prod_{q \nmid R} \frac{1 + q^{-\sigma}}{(1 - q^{-\sigma})^2} \leq \frac{\zeta(\sigma)^3}{\zeta(2\sigma)}. \quad (\text{D1})$$

Uniqueness of the prime-factorization indices and the coefficient bijections in (P1) justify the exact Euler products. In particular no success indicator has been inserted before constructing the series.

For a Dirichlet character $\chi \bmod R$, including imprimitive ones, put $\zeta_R(s) = \prod_{q \nmid R} (1 - q^{-s})^{-1}$. Applying χ to the finite group coordinate gives

$$\chi(\mathcal{F}_E(s)) = \frac{\zeta_R(s) L(s, \chi) L(s, \chi^2)}{L(2s, \chi^2)}, \quad \chi(\mathcal{F}_M(s)) = \frac{\zeta_R(s)}{\zeta_R(2s)} L(s, \chi) L(s, \chi^{-1}). \quad (\text{D2})$$

For E, rewrite $1 + \chi(q)z = (1 - \chi(q)^2 z^2)/(1 - \chi(q)z)$ in (P1). For M use $1 + z = (1 - z^2)/(1 - z)$. These prove (D2) prime by prime in the absolutely convergent half-plane. The L -function conventions and their Euler products are standard [6, 25.15.1–25.15.4]; no primitive replacement or missing Euler factor at a prime dividing R is permitted. Finite character inversion gives the generating series of actual target counts by $|\Gamma_R|^{-1} \sum_{\chi} \overline{\chi(t_c)} \chi(\mathcal{F}_c)$. This amplitude observation retains the full original series as its source; it is not a support-faithful inverse by (S4).

There is an explicit uniform-in- R tail. Prime-exponent comparison gives $\tau(a^2) \leq d_3(a)$, since $2e + 1 \leq (e + 1)(e + 2)/2$. Counting ordered factorizations yields $\sum_{a \leq T} d_3(a) \leq T(1 + \log T)^2$. Writing $a^{-\sigma} = \sigma \int_a^\infty t^{-\sigma-1} dt$, summing nonnegative terms, and dropping the subtracted initial count gives, for $X \geq 1$,

$$\sum_{a > X} \tau(a^2) a^{-\sigma} \leq \sigma X^{1-\sigma} \left[\frac{(1 + \log X)^2}{\sigma - 1} + \frac{2(1 + \log X)}{(\sigma - 1)^2} + \frac{2}{(\sigma - 1)^3} \right]. \quad (\text{D3})$$

This proves the claimed convergence and explicit tail. The Euler carrier uses all positive a coprime to R , including its finite nonphysical prefix. The actual first-half positive-shell carrier is its restriction to $a > R/2$: in the ordinary amplitude series one subtracts the explicitly retained finite

Dirichlet polynomial $\sum_{a \leq R/2, (a,R)=1} P_c(a) a^{-s}$. In the split carrier this is restriction of the index set with the removed prefix recorded, not cancellation relabelled as absence. The result concerns the full infinite arithmetic array at fixed modulus, not a nonzero coefficient at every index. Nor is it the Gamma/period kernel of the other programme. Imposing that $4a - R$ is prime changes the coefficient subsequence and does not preserve the asserted Euler product without another argument.

8 Verification, provenance and the remaining pointwise obligation

The general inverse obstruction, positive coefficient maps, periodic formulas, capacity reduction, two-class theorem and Dirichlet identities have the proofs above. The 174-pattern theorem has the stated exhaustive finite certificate, checked by two independently implemented algorithms; this is not independent mathematical review.

The full residual-27 classification was also applied to all 4519 primes at most two million in the six stipulated hard classes modulo 840. It matched actual counts in every instance: 1214 occupied residual-27 shells, 1880 E states and 1800 M states. This is not new prime coverage. An independent small-modulus check includes all first-half shells of 44 primes 1 mod 4 through 500, with 80032 canonical candidates and 1043 hits. Every output witness is checked through (A4). No assertion about other residuals is inferred from these tables.

For a hypothetical false prime, the profile of $a = (p + 27)/4$ must avoid all 174 exact patterns. Analogous fixed-residual classifications can be generated from (P2), but $R = 27$ alone is not a universal selector. The next arithmetic step must connect different shells of the *same* p , retaining

$$a_{R+4t} = a_R + t, \quad \gcd(a_R, a_R + t) = \gcd(a_R, t).$$

Neither the validity of these identities nor an invertible ordinary Fourier matrix proves that one of those correlated profiles is occupied. The work here stops before that implication. It replaces further defect-based repackaging by actual occurrence-level arithmetic and identifies exactly why support activation and amplitude inversion cannot supply the implication on their own. No historical-priority, Lean-build, global ES, or RH claim is made.

References

- [1] Split-Zero research programme (2026). *SplitZero: Reconstruction, internal homology, and balanced finite jets*. `formal/splitzero/DERIVED_MATHEMATICS.md`, §§1–4, Git revision 98c3c5f287231aa4f42cba19a7ab827ae3afffae. Original support reconstruction and quotient conventions; source verification reports are not re-certified by the present calculation.
- [2] User-directed Split-Zero programme (2026). *Globalization Semiring Analysis*, maintained source edition, Theorem 6.1 and §6.2, source lines 6780–6949. The mixed-support monad separates absent faces, supported zero residues and supported infinitesimal amplitudes. Protected source text is not redistributed.
- [3] Clankers (2026, May). *Support-Completed Center Sectors, Koide Null Forms, and Fourier-Zero Removal in Residual Erdős–Straus Shells*. Maintained May 2026 source edition, Abstract and “The quotient target has no Fourier blind spot.” The full Fourier support is an assertion in the ordinary complex group algebra.
- [4] User-directed ES programme (2026, September 11). *Boolean product formulation*, maintained research note, 12 September 2026. Section 2, source lines 8330–8468, records the empty $p = 3361, a = 847, R = 27$ shell and a different successful shell at the same prime. This antecedent is not newly discovered here.

- [5] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *Journal of the Australian Mathematical Society*, 94(1), 50–105. §2, Propositions 2.2 and 2.6; arXiv:1107.1010. <https://arxiv.org/abs/1107.1010>.
- [6] National Institute of Standards and Technology (2026). *Digital Library of Mathematical Functions*, §25.15(i), equations 25.15.1, 25.15.2 and 25.15.4. Dirichlet series, Euler products and retained imprimitive local factors. <https://dlmf.nist.gov/25.15>.